

Muyang Zheng

muyzheng@ucdavis.edu | Homepage | Google Scholar | LinkedIn

Education

University of California, Davis

Master of Science in Computer Science, GPA: 3.83/4.0

California, USA

Sept 2025 - May 2027

Hefei University of Technology

Bachelor of Engineering in Information Security

Anhui, China

Sept 2021 - June 2025

Publications

Open-ended Video Game Glitch Detection with Agentic Reasoning and Temporal Grounding (Accepted at *ACM Multimedia 2026*) [Preprint] [Code]

Apr 2026

Muyang Zheng, Tong Zhou, Geyang Wu, Zihao Lin, Haibo Wang, Lifu Huang*

MIST: Jailbreaking Black-box Large Language Models via Iterative Semantic Tuning (Under review) [Preprint] [Code]

June 2025

Muyang Zheng, Yuanzhi Yao*, Changting Lin, Caihong Kai, Yanxiang Chen, Zhiqian Liu

Professional Experience

Research Assistant, PLUM Lab at UC Davis – Advised by Dr. Lifu Huang

Sept 2025 – Present

- Built GLIDE, a **LangGraph-based multimodal agent** for open-ended video game glitch detection, with a five-stage pipeline for video scanning, iterative analysis, temporal grounding, and structured output generation.
- Implemented a **Planner–Executor–Reflector** agent loop with VLM-based visual QA, targeted region zoom-in, and SAM3 object tracking; designed multi-agent debate and confidence-based stopping for iterative glitch verification.
- Developed **bidirectional temporal grounding** to merge fragmented evidence and refine event boundaries, improving average F1 from **14.47% to 36.05%** and mIoU from **0.28 to 0.51** across six multimodal backbones.
- Built VIDEOGLITCHBENCH, a benchmark of **5,238 videos** across 120 games, using an **LLM-assisted annotation pipeline** with human verification to produce glitch descriptions and temporal annotations.

Research Intern, Binjiang Institute of Zhejiang University – Advised by Dr. Meng Han

Aug 2024 – Apr 2025

- Benchmarked in-house LLM inference with BitNet and LLMPerf, analyzing throughput, latency, and **10,000+ API requests** to identify performance bottlenecks.
- Implemented **20+ jailbreak attacks**, including GCG and AutoDAN, along with keyword-based refusal detection and LLM-as-judge evaluation for an LLM safety platform.
- Built an **automated red-teaming pipeline** for commercial LLM APIs and ran **1,000+ jailbreak evaluations** to analyze attack success rates and model robustness.

Open-source Projects

Monocular Depth Estimation Model Fine-tuning | PyTorch, LoRA | [Project link]

Nov 2025

- Fine-tuned MiDaS DPT-Large on NYU Depth V2 using **LoRA** ($r=8$, $\alpha=16$), building a PyTorch training and evaluation pipeline with only **3.26M trainable parameters (0.94%)**.
- Improved $\delta 1$ accuracy from **0.796 to 0.831**, while reducing AbsRel from **0.155 to 0.135** and RMSE from **0.529 to 0.447** over the zero-shot baseline.

LLM Jailbreak Evaluation Framework | Python, Flask, Vue, SQLite | [Project link]

May 2025

- Built a **full-stack LLM red-teaming platform** with Flask, Vue.js, and SQLite, integrating over 15 jailbreak attacks and automated evaluation across commercial LLM APIs.
- Implemented MIST, a black-box jailbreak method based on iterative semantic optimization, and built **automated attack analysis/visualization** pipelines; awarded Outstanding Graduation Project in Anhui Province.

Skills

Coding Languages: Python, Java, C++ , SQL, JavaScript, C

ML/AI: PyTorch, Hugging Face Transformers, LoRA/PEFT, Vision-Language Models, Agentic AI, RAG, SAM3

LLM Systems: LangGraph, Agent Harness, vLLM, OpenAI-compatible APIs, LLM Evaluation & Red-Teaming

Software & Tools: Flask, React, Vue.js, SQLite, Git, Linux

Honors

Third-class Scholarship & Merit Student in 2021-2022 Academic Year

Outstanding Member of the Student Union in 2021-2022 Academic Year